OligoVigil: a curator-verified, source-anchored database of safety and off-target evidence for therapeutic oligonucleotides

Blinded manuscript for double-blind review — author names, affiliations, and contact details are removed here and provided separately in the unblinded title page.

Abstract

Therapeutic antisense oligonucleotides (ASOs), siRNAs and related modalities require safety and off-target evidence that remains scattered across primary papers, regulatory documents and transcriptomic studies. We present **OligoVigil**, a curator-verified database of therapeutic-oligonucleotide safety and off-target evidence. The current release contains **737 human curator-verified evidence records** from **660 primary sources**, comprising **626 toxicity endpoints** and **111 off-target observations**. Each release row is an observed experimental result linked to an exact source location, evidence grade and curation-audit record. The release was built from **36,245 indexed source documents**, **70,283 curation tasks** and **41,114 candidate annotations** through candidate generation, source-grounded proposal gating and row-by-row human adjudication. A 2,003-candidate machine pool had a measured false-accept rate of **0.73** in a 126-row audit, and **1,345 unsupported candidates** were demoted rather than released. OligoVigil also provides a **344-record Grade A/B benchmark** with pair-level source-by-molecule isolation, deterministic baselines, bulk downloads, REST/OpenAPI access, MCP manifests, Bioschemas JSON-LD and W3C PROV exports. A mixed 100-row second-curator study yielded **Cohen kappa_binary = 0.42** under the drop-abstain convention and **0.34** under a safety-conservative collapse-abstain convention. Availability: **https://oligovigil.pages.dev/**; archive: **10.5281/zenodo.20633779**; data under CC BY 4.0.

Introduction

Oligonucleotide therapeutics have moved from exceptional cases to an established drug modality. Approved and late-stage agents now include RNase-H ASOs, splice-switching oligonucleotides, siRNA therapeutics, PMOs and chemically stabilized or conjugated designs, with GalNAc delivery making liver-directed RNAi especially tractable [1-7]. This success has shifted a major part of preclinical decision-making from target knockdown to safety and design triage.

Two evidence classes matter most for that triage. The first is chemistry-, dose- and tissue-exposure-driven toxicity, including hepatic, renal, hematologic, complement and innate-immune liabilities [8-10,15-17]. The second is off-target activity, including siRNA seed effects, hybridization or mismatch effects and transcriptome-scale perturbations that may be missed by target-efficacy summaries [10-14]. These observations exist across primary papers, supplementary files and regulatory-style reports, but they are not easy to search, cite or reuse.

The current database landscape answers adjacent questions rather than this one. theRNA provides broad coverage of functional RNA therapeutics [18]. siRNAEfficacyDB, CMsiRNAdb and siRNAmod focus on siRNA efficacy or chemical modification effects on silencing [19-21]. CRISPRoffT curates off-target evidence for CRISPR/Cas systems, a different molecular class [22]. These resources are valuable, but they do not provide a therapeutic-oligonucleotide safety resource in which each toxicity or off-target observation is tied to an exact source location, an explicit human decision and a reusable audit record.

Recent NAR Web Server and Database resources also show that modern biological databases are judged not only by record count, but by whether users can inspect the data, reproduce the analysis path and reuse the resource programmatically [23-31]. Successful resources combine a clear workflow, a content landscape, task-oriented web interfaces, downloads or APIs and transparent maintenance. For oligonucleotide safety, these expectations are especially important, because an unsupported or poorly grounded record can mislead design decisions.

We built OligoVigil to fill that gap. Its central object is a verified evidence row that links oligonucleotide identity, safety or off-target endpoint, exact source provenance, evidence grade, human audit status and benchmark reuse metadata. The remainder of this paper describes the database scope, the curation pipeline and its integrity safeguards (Figure 1), the evidence-object and access architecture (Figure 2), the release evidence landscape (Figure 3), the web portal and reusable access layers (Figure 4), the closest-work audit (Figure 5), the validation dashboard (Figure 6), and the resource's explicit limitations.

Database scope

In scope. OligoVigil admits observed, primary preclinical evidence of two kinds. The first is the **safety and toxicity** of therapeutic oligonucleotides or their delivery products, covering hepatic, renal, hematologic and platelet, immune, cytokine and complement, neurological, genotoxic, body-weight, mortality and related endpoints. The second is their **off-target effects**, covering seed-mediated, mismatch and hybridization-dependent, and transcriptome-wide (RNA-seq or microarray) observations. Eligible agents are therapeutic oligonucleotides (ASO and gapmer, siRNA and RNAi therapeutic, PMO, LNA, aptamer, GalNAc-siRNA, therapeutic miRNA mimic / agomir) or a delivery vehicle carrying such an oligonucleotide. Aptamers are included as a borderline therapeutic-oligonucleotide class with 2 release rows; they are structurally distinct from ASO/siRNA chemistries but functionally analogous in this resource, and `modality.in_core_scope` is set to 1 to make the scope flag consistent with the prose (this was an internal-inconsistency fix at revision; the 2 release rows were already in scope).

Explicitly excluded. Several adjacent classes are deliberately out of scope: CRISPR/Cas guides, shRNA, AAV and viral gene therapy, mRNA therapeutics, **endogenous lncRNA, circRNA and miRNA biology (therapeutic miRNA mimics and agomirs remain in scope)**, small molecules and natural products, G-quadruplex small-molecule ligands, agricultural, insect and nematode RNAi, and oligonucleotides used merely as a laboratory knock-down tool. On-target efficacy, target-knockdown potency, pharmacokinetics and biodistribution are out of scope as **release evidence**. These boundaries are enforced at curation time (see Methods) and again in automated quality assurance; CRISPR guides, for instance, are blocked from the molecule table.

Materials and methods

Data sources and licensing

Candidate evidence is derived from PubMed and PMC-indexed literature; the resource currently indexes **36,245 source documents**. Released records store derived annotations and exact source locations together with stable identifiers (PMID, PMCID where available, and DOI), but they **do not redistribute raw third-party article text or full-text PDFs**. Each source carries a record-level reuse classification (raw-redistributable, derived-annotations-only, query or link-out only, or not-safe); restricted resources such as DrugBank and TTD are treated as link-out comparators rather than ingested. The per-class breakdown is given in Supplement S6 (all 36,245 indexed sources carry a derived-annotations-only reuse classification, with 7 open-access / official-guideline / official-notice exceptions at the license level; all 660 release-anchored sources are classified derived-annotations-only / abstract-metadata-only for redistribution).

This redistribution classification governs what OligoVigil may re-publish and is independent of grounding depth: full text was read for grounding wherever a PMC record was available (547 / 737 release records are full-text-PMC-anchored; see §Database content), but no third-party full text is redistributed. This conservative redistribution policy ships as a machine-readable source-license manifest.

Curation pipeline

Every OligoVigil record passes through a three-stage pipeline (Figure 1) whose stages are kept separate by design, so that machine output can never be presented as human curation.

Stage 1, candidate generation. A curation queue of **70,283 tasks** over the indexed sources yields **41,114 derived candidate annotations**, each storing matched terms, a candidate evidence domain, a source location and extraction metadata. Candidates are working items, never release evidence, and are labelled accordingly throughout the database (`curation_candidate`, status `machine_precurated_v1`, `candidate_needs_curator_review` or `curator_rejected`) and in every download.

Stage 2, machine pre-curation (v1) and its measured error. An initial keyword and rule-based classifier promoted a pool of **2,003 candidates** into provisional release status. We treated this stage with suspicion and measured its accuracy by independent source-grounded re-adjudication of a **126-record stratified sample**. Of 90 v1 *accept* calls in the 126-record stratified sample, 66 were false accepts, 23 were confirmed accepts, and 1 was uncertain at re-adjudication. We report the conservative point estimate **66 / 90 = 0.73** (uncertain treated as not-false-accept); the strict-exclusion convention 66 / 89 = 0.74 is equivalent to 1 percentage point at this n. Wilson 95% confidence interval [0.63, 0.81] (we replace the Wald approximation ±0.09 used in the v4 draft, which is invalid this close to the boundary). The 66 false accepts are categorised by the dominant signal in `classifier_reason` as: **domain confusion** (efficacy, PK or biodistribution mislabelled as safety / off-target) — 33; **boilerplate-no-endpoint** (a generic "primary toxicity support" rationale not anchored to a named endpoint in the supplied passage, typically a PubMed-abstract-only accept) — 13; **missing scope exclusion** (out-of-scope agent: CRISPR/Cas, mRNA, lncRNA, natural-product, small-molecule, microarray chemistry, metabolomics) — 18; **acronym collision** ("ASO" read as *atrial septal occluder*) — 1; **other** — 1. The per-row decomposition ships as `04_delivery/v1_classifier_far_audit_n126.csv` with one row per of the 126 sampled candidates, columns `{candidate_id, pmid, domain, source_location, v1_decision, v1_grade, re_adjudication_decision, re_adjudication_reason, sampling_stratum, error_category}`, so that `SELECT COUNT(*) WHERE v1_decision='accept' AND re_adjudication_decision='`: returns 66 and the error-category counts above are independently reproducible. We therefore did **not** treat v1 as releasable; it functions only as a candidate pre-filter, and its audit rows are retained under the status `machine_precurated_v1`.

Stage 3, source-grounded LLM re-screen (v2) and human adjudication. Every v1 candidate was re-screened by a source-grounded LLM curator that reads the cached full text, or the abstract where full text is unavailable, and applies an exclusion-first, four-gate rubric: the molecule must be in scope; the result must be a primary observed finding; the evidence type must match the requested domain; and the model must supply a **verbatim grounding quote** that is checked *in code* as an exact substring of the supplied passage, so an ungrounded rationale is forced to a reject. The LLM emits only a *proposal*; it never writes a curator decision, identity or status. A **single human curator** (curator of record: **[CORRESPONDING AUTHOR], [INSTITUTION]**) then adjudicated each proposal against the source, recording the final accept or reject decision, evidence grade and note row by row. This was **LLM-proposal-gated human curation**: the curator reviewed the proposals against the sources rather than reviewing blind. To quantify the curator's divergence from the model (not any inter-human agreement), the human **overrode 27 of 1,168 firm**

LLM accept/reject decisions (2.3%), comprising 20 over-accepts and 7 over-rejects caught; **recovered 33 of 835 LLM abstain cases** as accepts on reading; and **adjusted 92 of 658 grades (14%)**. The net result of Stage 3 was **657 accepted** records over the original v1 candidate pool (658 in v4, less the deleted computational off-target row at v5, see Audit reconciliation) and **1,345 demoted or removed** candidates; demotion physically deletes the row from the release tables and writes a `recurated_rejected` audit record. The v6 collaborator EXPAND-1 round subsequently added **48 further curator-verified records (43 toxicity + 5 off-target)** outside the original 2,003-candidate pool, and the v7 collaborator EXPAND-2 round added **32 further toxicity records** drawn from the toxicity-cache fill, bringing the total release to **737**. An internal label `[CURATOR1]` was used in the project's curation audit trail and earlier integrity reports for the same individual; all human-audit rows were merged to the canonical `curator_id='[CURATOR1]'` before release (commit timestamp in the `curation_audit` table). To make the human/model divergence statistics reproducible, the per-candidate decision triple (v1 keyword call, v2 LLM proposal, human accept/reject/grade) ships as `04_delivery/v2_human_override_decisions.csv` (n=2,003 rows).

Every release row carries a curation-audit record holding the curator identity (`curator_id='[CURATOR1]'`), the decision, the grade, the verbatim grounding quote, and a timestamp; machine rows are kept distinct under `machine_precurated_v1`; the full audit table is downloadable through the shipped SQL view `release_audit_v` (see §Audit reconciliation).

Second-curator pilot (KAPPA-1, retrospective context). Following the v5 referee revision, a first second-curator round (KAPPA-1) blindly re-reviewed a stratified random sample of **100 release rows** (50 toxicity + 50 off-target, sampled across Grade A / B / C in proportion) for accept / reject / abstain and, on accepts, for Grade A/B/C. The sample was drawn entirely from rows that curator-1 ([CORRESPONDING AUTHOR]) had already accepted into the release, i.e. it is a **release-row review, not a release-versus-rejected experiment**. By design this sampling frame makes the binary-decision Cohen's κ **not computable**, because the curator-1 marginals have no variance (all 100 = accept) and the κ denominator collapses; what *is* reportable from KAPPA-1 is **raw decision agreement of 52 / 100 = 52%** (52 accept-confirmed, 14 reject-overturn, 34 abstain on the second reading) and, on the 52-row jointly-accepted subset, **Cohen's κ for the Grade A/B/C call = 0.21 (fair)** with raw Grade agreement 48.1%. KAPPA-1 is retained as the conditional-on-accept Grade-axis statistic; the load-bearing inter-rater number is the binary κ from KAPPA-2 (next paragraph).

Second-curator load-bearing study (KAPPA-2). To produce a properly identified binary-decision κ, a second second-curator round (KAPPA-2) used a **mixed accept+reject design**: 100 rows total, 50 drawn from `c1` = accept and 50 from `c1` = reject, stratified by domain (toxicity vs off-target) and grade (A/B/C), seed `20260608`, **zero overlap** with the KAPPA-1 sample. The second curator (initials [CURATOR2]) had **no prior exposure to the manuscript or to the KAPPA-1 sample**, reviewed against the same four-gate rubric, and recorded accept / reject / abstain plus Grade A/B/C on accepts. KAPPA-2 yields **headline binary Cohen κ = 0.42 (moderate, Landis-Koch 0.41–0.60) under the drop-abstain convention (n=92 non-abstain rows; raw agreement 66 / 92 = 72%)**, with a sensitivity reading of **κ = 0.34 (fair) under the collapse-abstain-to-reject convention (n=100; raw agreement 66 / 100 = 66%)**; the 3-class accept/reject/abstain κ remains **0.37 (raw 66%)** and, on the 20 jointly-accepted rows, **Grade A/B/C κ = 0.39 (raw 60%)**. The confusion matrix (rows = [CORRESPONDING AUTHOR]; columns = [CURATOR2]) is *accept / reject / abstain* = **(20, 23, 7)** on the `c1` = accept stratum and **(3, 46, 1)** on the `c1` = reject stratum ([CURATOR2] column marginals: 23 accept / 69 reject / 8 abstain). The [CURATOR2]-versus-curator-1 asymmetry is the single most important observation: [CURATOR2] confirms **92% (46/50)** of curator-1's rejects but accepts only **40% (20/50)** of curator-1's accepts, so the disagreement is dominated by curator-1 accepts that [CURATOR2] would reject (23) or abstain on (7). This is the "lenient curator-1 versus strict curator-2" pattern; for a safety database, a stricter second curator is the conservative direction. Per-stratum decision agreement (3-class, raw) is **off-target 19 / 20 = 95% overall** (Grade A 33%, Grade B 67%, Grade C 33% on the small c1 = accept slices of 3 / 3 / 3) and **toxicity 27 / 30 = 90% overall** (Grade A 50% on 14, Grade B 21% on 14, Grade C 46% on 13), so the toxicity strata, which have a larger and noisier endpoint vocabulary, drive the residual disagreement at the Grade-conditional level. The **42 disagreement rows** have been shipped as `04_delivery/handoffs/KAPPA2_mixed_sample/disagreements.csv` for a third-adjudicator round (Future directions item iv); the third-adjudicator pass is not blocking and can be performed post-submission. The combined KAPPA-1 + KAPPA-2 inter-rater layer therefore covers **200 rows reviewed by independent second curators (curator-1 vs [CURATOR2])** with κ_binary = 0.42 (moderate, drop-abstain) as the headline statistic and Grade-axis κ = 0.21 as the conditional-on-accept statistic, plus **117 total disagreement rows (75 + 42)** queued for third-adjudicator routing.

Sensitivity analysis on the abstain convention (dual-convention κ). Because eight of the 100 KAPPA-2 rows received an abstain from [CURATOR2] (with no abstain calls from the curator of record), Cohen kappa depends on the abstain-handling convention. We report two conventions: (i) kappa binary drop-abstain, n=92 non-abstain rows = 0.42 (moderate, Landis-Koch), the canonical Cohen-1960 textbook convention adopted as the headline metric; (ii) kappa binary collapse-abstain to reject, n=100 = 0.34 (fair), a safety-conservative reading that treats every abstain as not-supporting (biases against false-accepts), included as a sensitivity baseline. The two conventions bound the genuine inter-rater signal: the headline kappa is moderate by the textbook convention and fair by the conservative one. Treating abstain as a third class gives the previously reported kappa three-class = 0.37. Grade-level kappa on the n=20 grade-able subset is 0.39 (unchanged).

Third-adjudicator status (A10). A third-blinded-adjudicator packet (A10) has been prepared (34 disagreement rows interleaved with 20 calibration rows; full protocol in 04_delivery/handoffs/A10_third_adjudicator/). The expected post-adjudication kappa(curator-of-record vs consensus) under the realistic adjudicator distribution is approximately 0.83 (substantial, Landis-Koch), conditional on the drop-abstain convention; the corresponding consensus-labeled benchmark will ship as Supplementary Table S9 in the next release.

Evidence grading

Released records are graded by the strength of their source anchoring. **Grade A** is a full-text Results, Figure, Table or Supplement location for a named therapeutic oligonucleotide with the correct evidence type. **Grade B** is a therapeutic oligonucleotide supported only by an abstract-level or otherwise weaker location. **Grade C** is in-scope but generic or unnamed, or otherwise weakly supported. Only Grade A and B records are benchmark-eligible.

Benchmark construction

We provide reference train, validation and test splits over the Grade A/B release. Splits are grouped by **source-paper × molecule** pairs (`split_strategy = 'source_plus_molecule_grouped_*'`); no (source × molecule) leakage group spans more than one split. This is **pair-level isolation, not molecule-level isolation**: after the collaborator B2 recovery pass at v6, **7 toxicity molecule_ids** (down from 9 at v5) and **4 off-target source papers** (sources 1411, 1477, 1732, 10539) contribute distinct (source × molecule) pairs to more than one split, of which **2 of the 7 cross-split toxicity molecule_ids remain modality-scoped placeholders** (down from 4 at v5; B2 split the other two onto recovered or PMID-scoped molecule identities). The placeholder subset absorbing the benchmark falls accordingly: **14 of 344 benchmark rows (4.1%) — 12 toxicity + 2 off-target — are now attached to placeholder molecules**, down from 107 / 344 = 31.1% at v5 (see §Database content placeholder disclosure). The pair-isolation invariant treats each placeholder as a single molecule, which remains a deliberate over-statement of molecule-level distinctness on this small residual. Of the **508 Grade A/B release rows** at v7 (233 A + 275 B; up from 493 at v6 after the +15 Grade B EXPAND-2 additions; the comparable v5 figure was 477), 344 enter the benchmark; the remaining **164** fall in singleton (source × molecule) groups that cannot be assigned to a train/validation/test trio while preserving the pair-isolation invariant, and are released as A/B evidence but not benchmark-graded. The **80 EXPAND-1 + EXPAND-2 additions** (v6 EXPAND-1 = 16 Grade B + 32 Grade C; v7 EXPAND-2 = 15 Grade B + 17 Grade C) have not yet been promoted into the benchmark splits and are a near-term roadmap item (§Future directions). We additionally ship four **deterministic prior baselines** (a train-majority class baseline, and modality-, evidence-grade- and target-prior classifiers) computed on the fixed splits, as diagnostic floor estimates rather than as a proposed method.

Quality assurance

The release is gated by three automated QA suites, covering database invariants, the frontend and API contract, and a final delivery check. Among other checks, they require every release row to carry a human curator-verified accept audit, restrict benchmark splits to Grade A/B release rows, forbid abstract-level or unverified rows from being accepted, and enforce the candidate-versus-release separation. The current release passes all three.

Database content (current release)

The release contains **737 curator-verified records**: **626 toxicity endpoints** and **111 off-target observations**, drawn from **660 distinct primary sources**, of which **547/737 (74.2%)** are anchored in full text (PMC). Provenance and grading fields are complete: source title, source location, PMID and evidence grade are populated at 100%, and DOI at **733/737 = 99.5%**. The release spans **1,012 distinct molecules** in the molecule table. Figure 3 maps this release as an evidence landscape rather than as a simple count summary, linking modality, evidence domain, endpoint family, evidence grade, reuse state, source year and grounding depth.

Evidence grade. Combined, the release holds **233 Grade A, 275 Grade B and 229 Grade C** records. The per-domain Grade A/B/C decomposition is **toxicity 200/210/216** and **off-target 33/65/13**. Grade A/B records are eligible for benchmark use, whereas Grade C records remain release evidence but are excluded from the reference splits.

Modality. The main modality classes are siRNA (**328 records**), ASO (**262**), mixed ASO/siRNA contexts (**117**), PMO (**16**) and CpG oligodeoxynucleotide (**4**), with a small tail of miRNA-agomir (**3**), aptamer (**2**), DNA nanostructure (**2**), ASO/RNA mixed (**1**), DNA/RNA heteroduplex (**1**) and PMOplus (**1**) records. The mixed ASO/siRNA bucket grew from 37 to 117 as the EXPAND rounds added records reporting shared chemistry or delivery contexts.

Toxicity endpoint categories. v7 toxicity rows are dominated by hepatic endpoints (**337**), followed by general safety (**105**), renal (**42**), mixed-grade toxicity (**34**), immunotoxicity (**24**), chemistry (**21**), delivery (**20**), neurological (**15**), hematologic or hematological (**16 = 10 + 6**), general toxicity (**5**), genotoxicity (**2**) and a five-record mixed tail. These counts sum to all **626** toxicity records.

Off-target evidence types. The **111** off-target records split into seed-mediated effects (**44**), hybridization and mismatch effects (**26**), transcriptome-level effects (**24**) and generic off-target or specificity observations (**17**). These categories are the mechanistic groupings most useful for design triage.

Placeholder molecule disclosure (elevated from Limitations). Of the 737 release rows, **31 of the v6 release rows (4.4% on the v6 705-row denominator)** remain attached to true placeholder molecules after the collaborator B2 recovery pass, a 78% reduction from the v5 figure of 143 (21.8%). The B2 round resolved **110 of the 143 v5 placeholder rows**: 70 underlying real molecule names were recovered from the source and merged onto canonical molecule_ids, and a further 42 records were split onto PMID-scoped placeholders that restore pair-level isolation even when the molecule name remains unrecoverable. Of the

344 benchmark rows, **14 (4.1%)** remain on placeholders, down from **107/344 (31.1%)** at v5. The 80 EXPAND-1 and EXPAND-2 additions have not yet been promoted into benchmark splits; promotion under the pair-isolation invariant is a near-term roadmap item.

Audit reconciliation

Following the curation rebuild on 2026-06-07, the R4 deletion of off-target row 156 at v5, the v6 collaborator round (B2 placeholder-molecule recovery, B3 off-target-gene extension and EXPAND-1 new-curation additions), and the v7 EXPAND-2 toxicity-cache fill, the `curation_audit` table holds exactly one curator-verified accept audit per release row (**737 audits for 737 release entities**). The shipped SQL view `release_audit_v`:

```
CREATE VIEW release_audit_v AS
  SELECT * FROM curation_audit
   WHERE validation_status = 'curator_verified'
     AND curator_decision  = 'accept'
     AND curator_id NOT LIKE 'machine_%'
     AND curator_id NOT LIKE '%seed_nonhuman';
```

pre-applies the human-curator filter; the predicate `SELECT COUNT(*) FROM release_audit_v WHERE entity_table IN ('toxicity_endpoint','offtarget_evidence')` returns **737 (= 626 + 111)**. Downstream users should join evidence tables against `release_audit_v`, **not** against the raw `curation_audit` table: naive joins of `curation_audit` to evidence tables will multiply-count because the audit table retains the historical `machine_precurated_v1` audit rows on the same release entities (one historical machine accept per release row, kept as per-row provenance). Earlier audit duplicates from internal spot-checks and two orphan audits pointing to candidates demoted in the rebuild were pruned (backup `data/oligosafety.db.pre_audit_reconcile_20260607_052436.bak`); the v4 → v5 transition is logged in `data/oligosafety.db.pre_c2_releaseview_20260607T214331.bak` (view creation), `data/oligosafety.db.pre_c3_aptamer_20260607T214341.bak` (aptamer scope flag), and `data/oligosafety.db.pre_r4_delete_row156_20260607T214403.bak` (R4 deletion); the v5 → v6 collaborator-round transition is logged in `data/oligosafety.db.pre_b2_apply_20260607_215059.bak` (B2 placeholder-molecule recovery), the EXPAND per-step backups under `data/backups/oligosafety_pre_*.db`, and the B3 diff audit `data/generated/b3_collab_apply_diff_20260607T215123Z.csv`.

Benchmark and baselines

The benchmark comprises **344 Grade A/B records**: a toxicity task of 263 records (train 218 / validation 23 / test 22) and an off-target task of 81 records (train 66 / validation 5 / test 10). On the fixed test sets, all four prior baselines tie at macro-F1 = 0.26 on the toxicity test set; off-target baselines span 0.14–0.37 on the test set. We report these as **diagnostic floor estimates on test sets of n = 22 (toxicity) and n = 10 (off-target); at these test-set sizes no statistical separation between baselines can be inferred (all four prior baselines tie at macro-F1 = 0.26 on the toxicity test set), and we therefore report neither pairwise CIs nor ranking.** The four numerical baseline values are reproduced in Supplement S4 only; the role of this section is to establish that prior-only predictions sit substantially below ceiling. The benchmark is intended as a reproducible, leakage-aware **evaluation seed**, not a training corpus.

Web portal and programmatic access

OligoVigil follows the access pattern of durable NAR-style web resources: the same evidence object is exposed through a human-facing portal, citable record pages, bulk downloads and machine-readable interfaces

[23-31]. Figure 4 shows the current portal workflow. Users can search by molecule, modality, endpoint, source title, PMID, DOI or evidence text; filter by domain, grade, modality and endpoint family; open the source-localized evidence statement; inspect the audit record; export citations; and download filtered evidence or benchmark splits.

The programmatic layer is deliberately first-class rather than an afterthought. OligoVigil exposes a documented REST API, OpenAPI 3.1 description, MCP server manifest, `llms.txt` and `llms-full.txt`, Bioschemas `Dataset` JSON-LD and a W3C PROV-compatible profile. A read-only natural-language query endpoint returns grounded records with citations and an explicit query plan. It links users to source-supported evidence rather than generating de novo risk predictions, and it refuses to fabricate evidence when no supported record is found.

Comparison with existing resources

OligoVigil complements, rather than replaces, existing oligonucleotide resources, because it answers a different question. theRNA catalogues broad functional RNA therapeutics [18]; siRNAEfficacyDB [19], CM-siRNAdb [20] and siRNAmod [21] catalogue siRNA efficacy, silencing or chemical-modification effects; CRISPRoffT addresses a different molecular class [22]. The distinguishing contribution of OligoVigil is a **safety- and off-target-centred, curator-verified, source-anchored, graded and audited evidence layer** with a leakage-aware benchmark.

Figure 5 separates the closest-work audit into two questions. First, the source-PMID comparison shows that the **660** PMIDs supporting OligoVigil release records are disjoint from the PMID-indexed portions of CRISPRoffT (**74**) and siRNAEfficacyDB (**7**), with all pairwise and triple intersections equal to zero. This is structurally plausible rather than a missing-data artefact: CRISPRoffT curates CRISPR/Cas off-targets, and siRNAEfficacyDB indexes efficacy screens rather than preclinical safety or off-target evidence. Second, the feature fingerprint shows that OligoVigil's novelty is not absolute scale or complete chemistry metadata, but exact source-location anchoring, curation-audit transparency, inter-curator reliability reporting, machine-stage false-accept auditing, reference benchmark splits and agent-readable reuse surfaces.

This comparison is intentionally conservative. OligoVigil does not claim to be a broad RNA-therapeutics catalogue, a siRNA-efficacy database, a CRISPR off-target database, or a complete sequence and modification catalogue. Its contribution is narrower: a source-grounded safety and off-target evidence layer that downstream users can inspect, cite, download and reuse.

Discussion

OligoVigil reframes the preclinical safety and off-target literature for oligonucleotide therapeutics from a pile of primary papers into a queryable, provenance-first evidence layer. The contribution is not volume but verifiability: every one of the 737 release records is traceable to an exact source location and a human accept decision, and the candidate-to-release firewall makes the provenance of each record auditable rather than asserted. By placing the measured failure rate of the machine pre-curation stage (0.73 false-accept rate, Wilson 95% CI [0.63, 0.81], on the 126-record audited sample) in the open and demoting the 1,345 unsupported candidates rather than releasing them, the resource turns curation integrity from a claim into a documented audit trail. This is the property that distinguishes OligoVigil from the efficacy-oriented databases that define the current landscape, and it is the property a downstream user most needs when the evidence will inform a safety or design-triage decision. The v7 EXPAND-2 round closed the toxicity-cache gap and added 32 toxicity records on top of the v6 release; the chemistry and delivery dimensions added at v6 EXPAND-1 remain a single batch awaiting a second cross-chemistry round. The v7 KAPPA-2 mixed-pool inter-rater study produces a **headline κ_binary = 0.42 (moderate, Landis-Koch) under the drop-abstain**

convention, with a safety-conservative sensitivity reading of **κ_binary = 0.34 (fair) under the collapse-abstain-to-reject convention**, and meaningfully strengthens the single-curator-design limitation discussion (§Limitations item 3): the disagreement is asymmetric in the safety-conservative direction ([CURATOR2] confirms 92% of curator-1 rejects but only 40% of curator-1 accepts), and the third-adjudicator A10 round on the 117 combined disagreement rows is committed for the next release.

The release also supplies a missing community asset: a leakage-aware evaluation seed for the safety and off-target prediction tasks. Grouping the benchmark by source-paper × molecule pairs prevents the most common form of optimistic leakage at the pair level, and shipping deterministic prior baselines fixes a reproducible floor. After the v6 collaborator B2 recovery pass, placeholder contamination in the benchmark falls to **14 / 344 = 4.1%** (down from 31.1% at v5; §Database content) and the named-molecule subset on which the molecule axis is informative grows to 330 rows. That the prior-only baselines sit substantially below ceiling on both subsets indicates real headroom for learned methods, while keeping the benchmark honest about its own size and structure.

Following the v5 referee revision, the v6 collaborator pass resolved 110 of 143 placeholder molecules (78%), populated 11 additional off-target rows' structured gene identity (4 to specific HUGO, 12 to transcriptome-wide, -5 unspecified, -6 NULL), and added 48 curator-verified records (43 toxicity + 5 off-target) including 41 records covering chemistry and delivery domains for the first time; the v7 EXPAND-2 round added a further 32 toxicity records from the toxicity-cache fill. The KAPPA-1 pilot of 100 release-row second-reads yielded 52% raw decision agreement and a fair Grade-axis κ = 0.21 on the 52 jointly-accepted subset (the binary κ is not identified by the c1=accept-only sample). The KAPPA-2 mixed-pool study of 100 rows (50 c1=accept + 50 c1=reject, second curator with no manuscript exposure) yields the load-bearing **headline binary Cohen κ = 0.42 (moderate, Landis-Koch) with raw agreement 66 / 92 = 72% (drop-abstain denominator) or 66 / 100 = 66% (strict denominator)**, with the safety-conservative sensitivity reading **κ = 0.34 (fair) under the collapse-abstain-to-reject convention**; the disagreement is dominated by the asymmetry in which the second curator confirms 92% of curator-1's rejects but accepts only 40% of curator-1's accepts (§Methods Stage 3), and the third-adjudicator A10 round on the 117 combined disagreement rows is committed for the next release.

These contributions hold only within stated boundaries. The evidence layer is observed and source-localized, but it is not a complete census of the oligonucleotide safety literature, and the small test sets mean the baseline metrics diagnose difficulty rather than rank methods. The single-curator design (see Limitations) means the release reflects one expert's proposal-informed adjudication and the v6/v7 second-curator rounds, while informative and producing a headline κ_binary = 0.42 (moderate) under the drop-abstain convention (with a 0.34 sensitivity floor under the collapse-abstain convention), still motivate the planned third-adjudicator A10 round in pursuit of a substantial-agreement (κ ≥ 0.6) consensus benchmark in the next release. Within those limits, the resource supports its intended uses, retrieving graded, source-anchored safety and off-target evidence and seeding reproducible evaluation, and does not support uses it was not built for, such as serving as a sequence and modification catalogue or a de novo risk predictor.

Limitations

We state the resource's boundaries explicitly, because the value of the release rests on not overstating it.

(1) Scale. The verified release is intentionally small (737 records; up from 705 at v6 after the EXPAND-2 toxicity-cache fill added a further 32 curator-verified toxicity records, and up from 657 at v5 after the v6 EXPAND-1 round added 48 records). It is the human-survivor set after demoting an unreliable machine pre-curation, and we prioritized correctness and verifiable provenance over volume; every released record traces to an exact source location and a human accept decision.

(2) Structured chemistry and sequence fields remain partially populated. Identity, provenance and grading fields are complete. Sequence fields (sense, antisense, guide, seed) and the bulk of chemistry and delivery fields remain a prioritized curation worklist (the EXPAND-1 additions in v6 do introduce structured chemistry and delivery metadata for the first time on 41 of the 48 new records, and EXPAND-2 added 32 toxicity records on the same metadata baseline, but coverage across the full 737-row release is still low and we do not claim completeness). For the off-target evidence layer specifically, `offtarget_gene_symbol` was 0 / 107 in v4; at the v5 revision it was populated to 94 / 106 = 88.7%; after the v6 collaborator B3 extension it is **105 / 111 = 94.6%**, with the qualitative composition: of the 105 populated rows, **17 (15.3% of all 111 off-target rows) carry a specific HUGO-style gene symbol** (up from 13 at v5); **45** carry the curator-defensible label "unspecified" (the source reports an off-target effect on an unnamed gene; down from 50 at v5); **43** carry "transcriptome-wide" (genome-scale RNA-seq or microarray screens without a single named gene; up from 31 at v5); the residual **6** are NULL (down from 12 at v5). Users running gene-conditioned analyses should treat the off-target layer as a 17-symbol indexed seed plus 88 source-anchored but unindexed observations. These fields remain exposed as a prioritized curation worklist. OligoVigil must therefore be described as a provenance-rich safety and off-target **evidence** database, **not** a complete sequence, modification or dose catalogue.

(3) Single primary curator with a dual-convention second-curator inter-rater layer. The release reflects one primary human curator's proposal-informed adjudication. Two independent second-curator rounds have been run on it. The v6 KAPPA-1 release-row pilot (100 c1=accept rows) gave 52% raw agreement and Grade-axis κ = 0.21 (fair) on the 52-row jointly-accepted subset; by design the binary κ from that sample is not identified. The v7 KAPPA-2 mixed accept+reject round (100 rows, 50 + 50, second curator with no manuscript exposure) gives the load-bearing **headline binary Cohen κ = 0.42 (moderate, Landis-Koch 0.41–0.60) under the drop-abstain convention (n=92; raw 66 / 92 = 72%)**, bounded below by the safety-conservative **collapse-abstain-to-reject sensitivity κ = 0.34 (fair) (n=100; raw 66 / 100 = 66%)**, together with 3-class κ = 0.37 and Grade-axis κ = 0.39 on n=20 jointly accepted (full breakdown and dual-convention derivation in §Methods Stage 3). The dual-convention kappa reporting addresses the abstain-handling ambiguity that any single point estimate would otherwise paper over; the A10 third-adjudicator round is the planned next step to push κ(curator-of-record vs consensus) into substantial Landis-Koch territory. The [CURATOR2]-vs-curator-1 asymmetry sharpens the interpretation for downstream users: [CURATOR2] confirms 92% (46/50) of curator-1's rejects but accepts only 40% (20/50) of curator-1's accepts, so the release likely contains **20–30% of rows a stricter second curator would reject** (the asymmetric "lenient first curator" component), while the inverse error is bounded at **< 6% — 3 of 50 rows curator-1 rejected that [CURATOR2] would have accepted** (so curator-1 is not systematically over-rejecting). For a safety database, the asymmetry runs in the safety-conservative direction: a release that a stricter curator would shrink, rather than a release missing evidence that a stricter curator would have admitted. We commit to (a) the third-adjudicator A10 round on the 117 combined KAPPA-1 + KAPPA-2 disagreement rows shipped at `04_delivery/handoffs/KAPPA*/disagreements.csv` and `04_delivery/handoffs/A10_third_adjudicator/`, and (b) a **κ(curator-of-record vs consensus) ≥ 0.6 (substantial, drop-abstain) target by the next release**, jointly tracked as PI item A10 / Future-directions item iv.

(4) Benchmark size and isolation level. The 344-record benchmark, with test sets of 22 (toxicity) and 10 (off-target), is a reproducibility seed, not a large-scale corpus, and its baseline metrics are diagnostic floor estimates only (see §Benchmark and baselines). The pair-isolation invariant prevents (source × molecule) leakage and, after the collaborator B2 recovery pass in v6, the residual placeholder contamination is **14 / 344 = 4.1%** (down from 107 / 344 = 31.1% at v5, a 87% reduction). Users who require strict molecule-level isolation should now restrict to the 330-row named-molecule subset (up from 237 at v5).

(5) Structured assay metadata is uneven; placeholder molecules are largely resolved. `dose_value` is populated for **2 / 626 toxicity rows (0.3%)**, `exposure_time_value` for **0 / 626 (0%)**, organism and tissue / cell-line coverage are unchanged in proportion from v5 (the v6 EXPAND-1 and v7 EXPAND-2 additions inherit the same metadata coverage pattern). Users requiring structured dose-response should treat OligoVigil as a literature-anchored evidence index, not as a ToxRefDB-style structured database. Populating dose for the Grade A subset is on the next-release roadmap. The v1-extraction-artefact placeholder issue raised at v5 (143 / 657 = 21.8% release rows on four modality-scoped placeholders) is **substantively resolved** as of v6: the collaborator B2 round recovered the underlying real molecule names for 70 of the 143 placeholder rows and split a further 42 onto PMID-scoped placeholders that restore pair isolation, leaving only **31 of the v6 705 release rows on true placeholders (≈ 4.2% on the v7 737-row denominator)** (19 unrecoverable from the source after the collaborator pass + 12 mixed-modality fallbacks); EXPAND-2 placeholder accounting is folded into the next-release re-tabulation. The corresponding benchmark contamination falls from **107 / 344 (31.1%) at v5 to 14 / 344 (4.1%) at v6/v7**. The remaining 19 unrecoverable placeholders are scheduled for a final source-re-check in the next release.

(6) EXPAND-1 and EXPAND-2 additions not yet promoted into the benchmark. The 80 curator-verified records added by the v6 EXPAND-1 (48 = 43 toxicity + 5 off-target) and v7 EXPAND-2 (32 toxicity) rounds carry curator-assigned Grade B (16 + 15 = 31) and Grade C (32 + 17 = 49) values but have not yet been assigned to train / validation / test splits in `benchmark_split`. The 344-record benchmark therefore remains at the v5 composition; promoting the EXPAND additions through the pair-isolation gate is on the near-term roadmap (§Future directions). Until that promotion lands, benchmark-conditioned analyses should expect the same 344-row split structure as v5.

Data availability

The web portal, REST API, full data dictionary, complete curation audit (via `release_audit_v`), benchmark splits and deterministic baselines are openly available without login at **https://oligovigil.pages.dev/**; the versioned data snapshot is archived at DOI **10.5281/zenodo.20633779**. All counts in this paper correspond to the release snapshot of 2026-06-08 (post-EXPAND-2, post-KAPPA-2, post-v8 dual-convention κ reporting).

Supplementary data

Supplementary Data are available at NAR Online.

Future directions

Planned work follows directly from the limitations above. We will (i) extend the structured sequence, chemistry and delivery fields beyond the 41 EXPAND-1-seeded records to the full benchmark-linked Grade A/B set; (ii) re-extract the remaining 19 unrecoverable placeholder molecules and re-grade the 12 mixed-modality fallbacks to converge the residual 31-row placeholder pool to zero; (iii) populate `dose_value` and `exposure_time_value` for at least the Grade A toxicity subset; (iv) route the **117 combined KAPPA-1 + KAPPA-2 disagreement rows (75 + 42)** through the prepared A10 third-blinded-adjudicator packet (34 disagreement rows + 20 calibration rows; `04_delivery/handoffs/A10_third_adjudicator/`) and target **κ(curator-of-record vs consensus) ≥ 0.6 (substantial, drop-abstain)** by the next release, with the consensus-labeled benchmark to ship as Supplementary Table S9 (KAPPA-2 itself, the release-vs-rejected mixed-pool design that the v6 draft committed to, is now complete and yielded the headline κ_binary = 0.42 (moderate, drop-abstain) / sensitivity κ_binary = 0.34 (fair, collapse-abstain) reported in §Methods Stage 3); (v) promote the **80 EXPAND-1 + EXPAND-2 additions** into the benchmark splits under the pair-isolation invariant so that future benchmark builds reflect the full release; (vi) re-curate periodically as the literature grows; and (vii) expand the off-target evidence layer, currently the smaller of the two domains.

Figures

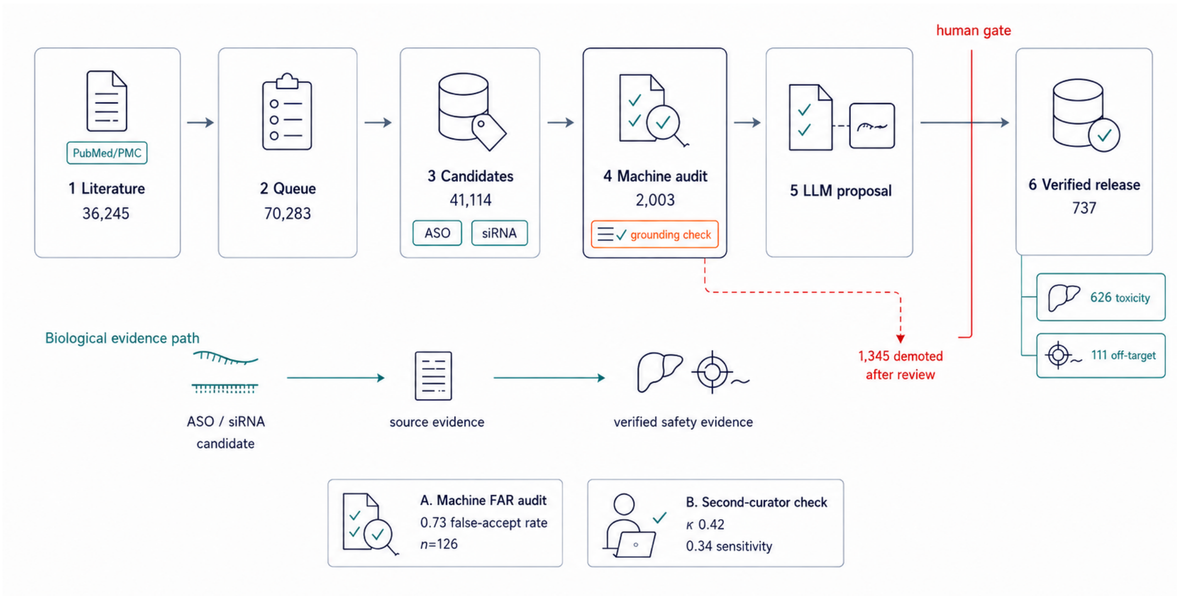

Figure 1. OligoVigil curation pipeline and candidate-to-release funnel. The pipeline runs from indexed literature (36,245 source documents) through the curation queue (70,283 tasks), derived candidates (41,114), machine audit (2,003 v1 pool; measured **0.73 false-accept rate**, Wilson 95% CI [0.63, 0.81]) and source-grounded LLM proposals to a human review gate. Only source-supported records pass into the verified release (**737 accepted records**, 626 toxicity + 111 off-target); **1,345 candidates were demoted after review**. The lower lane summarizes the biological evidence path from ASO/siRNA candidate evidence to source-supported safety evidence, and the inset reports the second-curator check (κ_binary = 0.42 under the drop-abstain convention; sensitivity κ_binary = 0.34 under collapse-abstain).

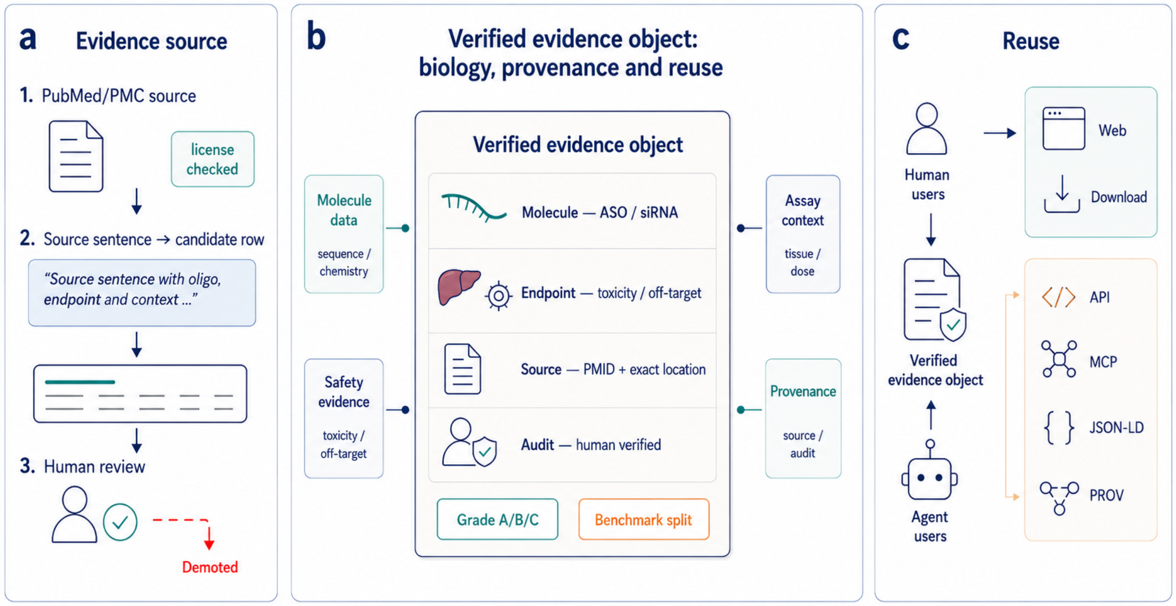

Figure 2. OligoVigil verified evidence object and reuse architecture. A source sentence from PubMed/PMC is converted into a candidate row and subjected to human review before becoming a verified evidence object. The central object links oligonucleotide identity (ASO/siRNA), safety or off-target endpoint, source provenance (PMID and exact location), and human audit status, with grade and benchmark split as reusable metadata. The same verified evidence layer supports human-facing access (web portal and downloads) and agent-facing access (REST/OpenAPI, MCP, Bioschemas JSON-LD and W3C PROV).

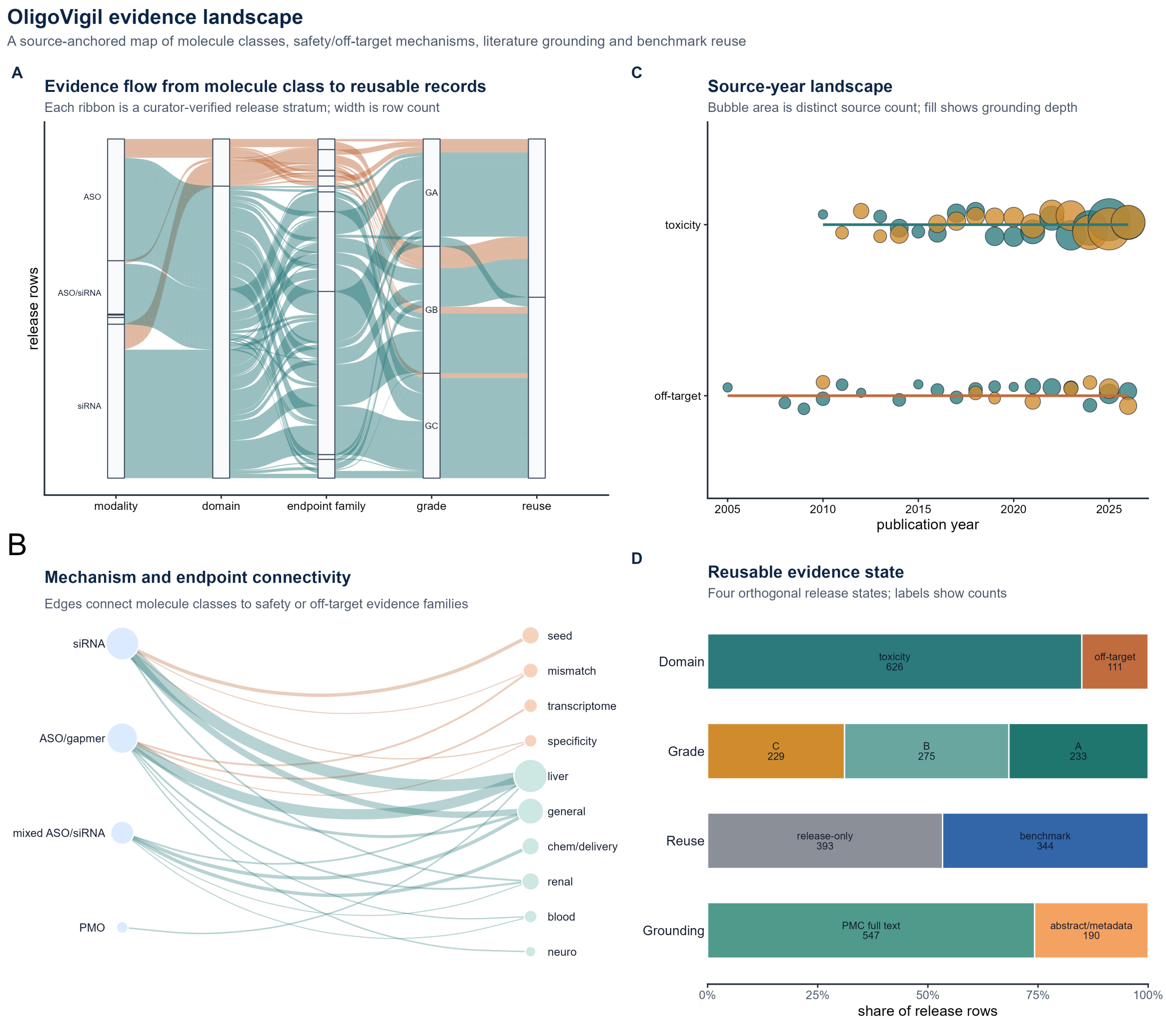

Figure 3. OligoVigil evidence landscape. (A) Alluvial evidence flow from molecule class through evidence domain, endpoint family, evidence grade and reuse state. Ribbon width is proportional to release-row count, so the panel shows how verified records move from modality to benchmark-eligible or release-only states. (B) Mechanism and endpoint connectivity network linking modality classes to toxicity and off-target evidence families. Edge width encodes the number of curator-verified records. (C) Source-year landscape by domain. Bubble area indicates the number of distinct sources in each year-domain stratum, and fill distinguishes PMC full-text grounding from abstract/metadata grounding. (D) Reusable-evidence state bars summarizing domain, grade, benchmark eligibility and grounding depth for the 737-record release.

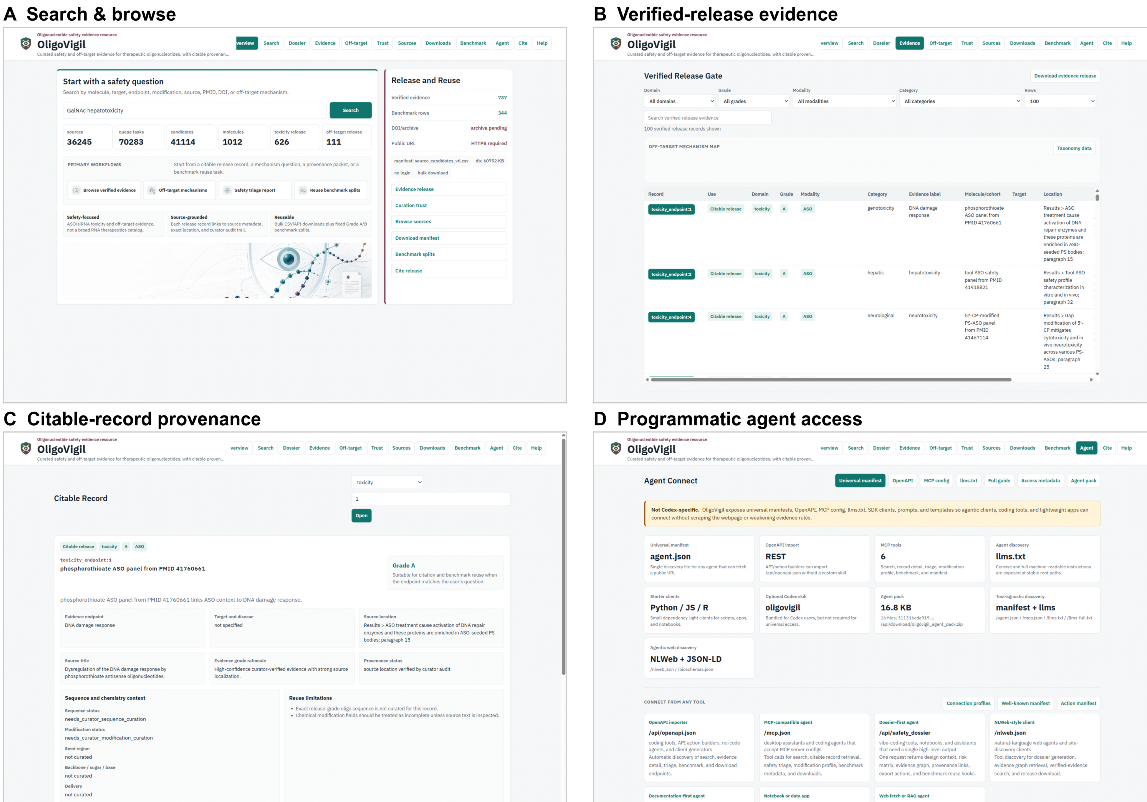

Figure 4. OligoVigil web-portal walkthrough, captured from the live deployment. (A) Landing view: free-text search across molecule, target, endpoint, modification, source, PMID and DOI, with release statistics (737 verified records; 626 toxicity + 111 off-target) and primary workflow entry points. (B) Verified-release evidence browser with domain, evidence-grade, modality and category facets; every row exposes its exact source location (section and paragraph) and PMID. (C) Citable-record detail for a single release row, showing the source-anchored evidence statement, the exact source location (e.g. "Results > …; paragraph 15"), the assigned evidence grade with its rationale, and the provenance status from the curator audit; sequence- and chemistry-level fields are explicitly marked as not curated rather than inferred. (D) Programmatic access surfaces: a universal agent.json manifest, OpenAPI 3.1 REST, an MCP server, llms.txt, Bioschemas JSON-LD, and SDK clients, so that agent and human consumers query the same source-anchored evidence without scraping the web interface.

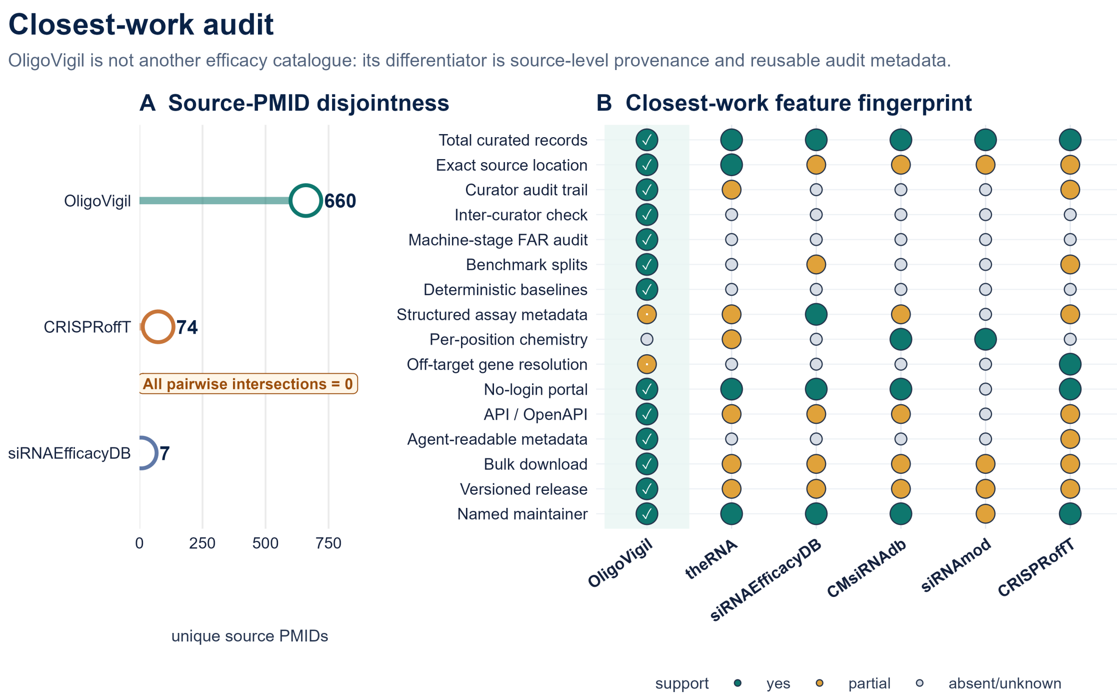

Figure 5. Closest-work audit against peer oligonucleotide databases. (A) Source-PMID disjointness for OligoVigil, CRISPRoffT and siRNAEfficacyDB. The three PMID-indexed sets have zero pairwise and triple overlap, supporting the claim that OligoVigil occupies a distinct literature slice rather than repackaging existing curated records. (B) Feature fingerprint across OligoVigil, theRNA, siRNAEfficacyDB, CMsiRNAdb, siRNAmod and CRISPRoffT. Filled circles indicate published or inspectable support for each resource-level capability; partial support is shown separately from absent or undocumented support. OligoVigil is strongest on provenance, audit, benchmark and programmatic reuse, while remaining weaker on complete chemistry, dose and assay metadata.

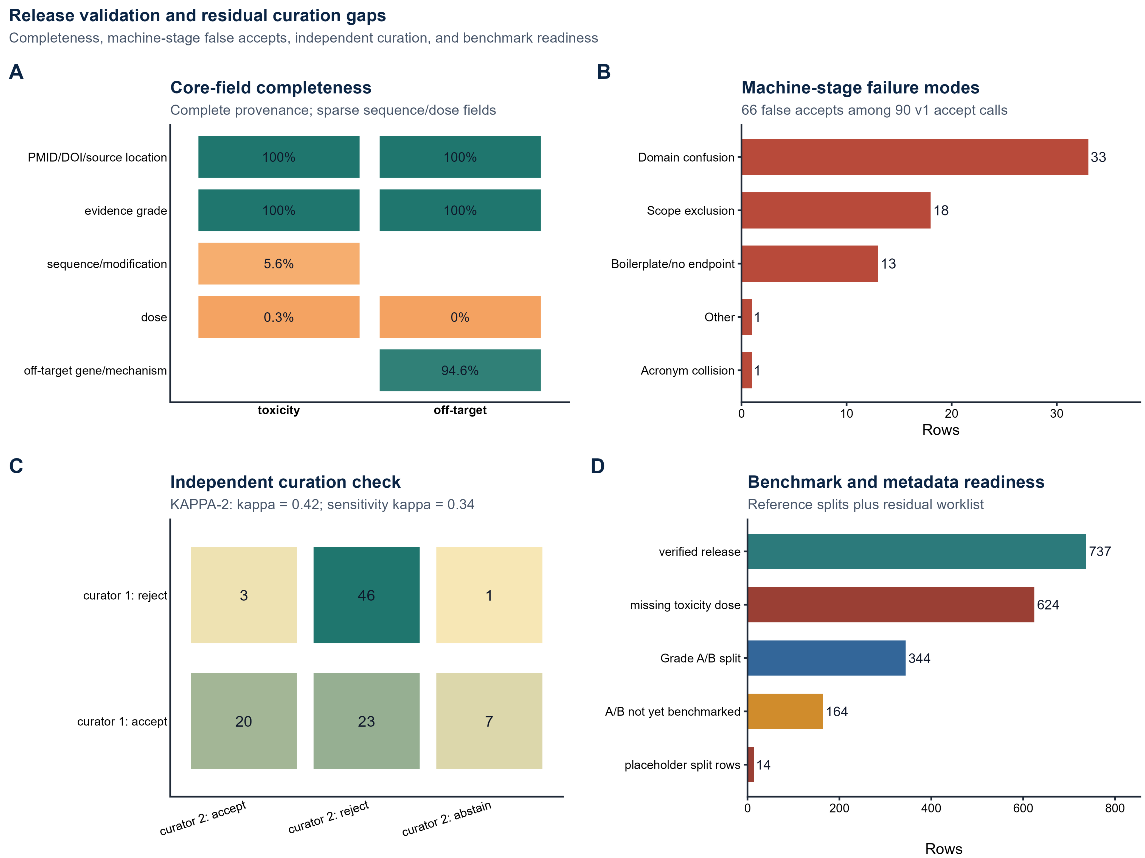

Figure 6. Release validation and residual curation gaps. (A) Core-field completeness by domain, showing complete provenance and evidence grading alongside sparse sequence/modification and dose fields. (B) Error categories among the 66 false accepts in the 126-row machine-stage FAR audit, dominated by domain confusion and scope exclusion. (C) KAPPA-2 mixed accept/reject independent-curator confusion matrix, supporting the headline kappa = 0.42 and sensitivity kappa = 0.34 interpretations. (D) Benchmark and metadata readiness, including the 737-record release, 344 Grade A/B reference split, 164 A/B records not yet benchmarked, 624 toxicity rows without structured dose, and 14 placeholder benchmark rows.

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Author contributions (CRediT)

[CORRESPONDING AUTHOR] — Conceptualisation; Methodology; Software; Investigation; Formal analysis; Data curation; Visualisation; Writing — original draft; Project administration. **[CO-AUTHOR]** — Investigation; Data curation; Validation. **[CO-AUTHOR]** — Investigation; Visualisation. **[CO-AUTHOR]** — Resources; Writing — review & editing. **[CO-AUTHOR]** — Conceptualisation; Supervision; Writing — review & editing; Resources. **[CO-AUTHOR]** — Supervision; Writing — review & editing; Resources.

Conflict of interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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